GGG-P-501e
August 26, 1963

SUPERSEDING
Int. Fed. Spec. GGG-P-00501d (Navy-Ships)
March 30, 1960 and
Fed. Spec. GGG-P-501c
February 5, 1953

FEDERAL SPECIFICATION

PLUMB BOB

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plumb bobs commonly used by surveyors, builders, and machinists for establishing perpendicular lines and for locating any point vertically above or below another point.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity indicated by the title of the specification, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* Plumb bobs shall be of the following types and classes as specified (see 6.2):

Type I.—Brass, with replaceable points.
Type II.—Iron, cast.
Type III.—Steel.
  Class A.—Mercury-filled, with replaceable points.
  Class B.—Solid.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of plumb bobs otherwise conforming to this specification.

3.2 Material. Material shall be as specified herein for each type.

3.3 Identification marking. Each plumb bob shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. The marking on the points for type I plumb bobs shall be optional with the manufacturer.

3.4 Cord hole or slot. The hole provided for the cord in all plumb bobs shall be either chamfered, or the edges free of burrs to assure proper rotation for the concentricity test of 4.4.1. When a slot is provided, the slot shall center the cord with the longitudinal axis of the plumb bobs, and also be chamfered for conducting the concentricity test in 4.4.1.

3.5 Threads. All threaded parts shall be of a nationally recognized commercial standard thread having standard size, shape, and tolerances in order to insure correct mating of parts and convenience of interchangeability and replacement.

3.6 Type I brass plumb bobs (with replaceable points. Type I plumb bobs shall consist essentially of a body, a removable head, and a replaceable point. A minimum of 7 feet of braided nylon or linen cord, having a tensile strength of not less than 10 pounds, shall be provided with each plumb bob. Type I plumb bobs shall be similar to A or B of figure 1 and shall conform to table I.

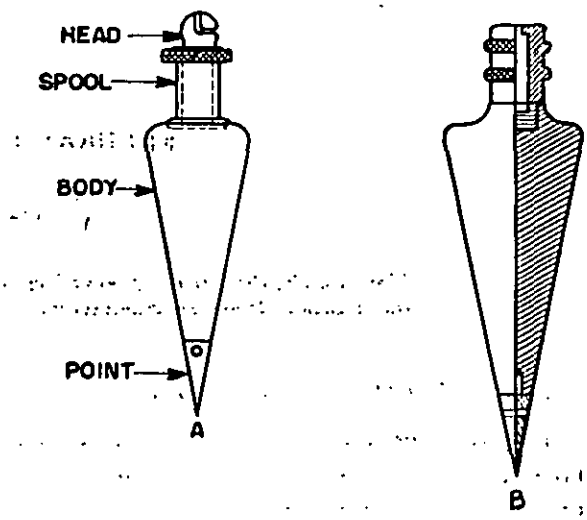


FIGURE 1.—Type I plumb bob, brass, with replaceable points

Table I - Type I plumb bobs, brass, with replaceable points.

Size weight nominal	Overall length		Diameter of bob at largest section	
	Maximum	Minimum	Maximum	Minimum
Ounces	Inches	Inches	Inches	Inches
6	5-1/4	4	1-3/16	1
8	5-3/4	4-3/8	1-1/4	1-1/16
10	6	4-3/8	1-5/16	1-3/16
12	6-1/8	4-7/8	1-3/8	1-1/4
14	6-1/4	5-1/4	1-1/2	1-3/8
16	6-1/4	5-3/8	1-5/8	1-3/8
18	6-1/2	5-1/2	1-3/4	1-1/2
24	7	6-1/8	1-7/8	1-1/2
32	7-3/4	6-3/4	2-1/8	1-13/16

3.6.1 *Body.* The plumb bob body shall be a machined conical brass body made from rolled or forged barstock to insure a homogeneous body free from blowholes and shrink holes. The body shall have an included angle of taper between 19 and 24 degrees. The upper portion of the body shall be threaded to receive the removable head. The lower portion of the body shall be machined to receive the replaceable point in accordance with the dimensions shown on figure 2.

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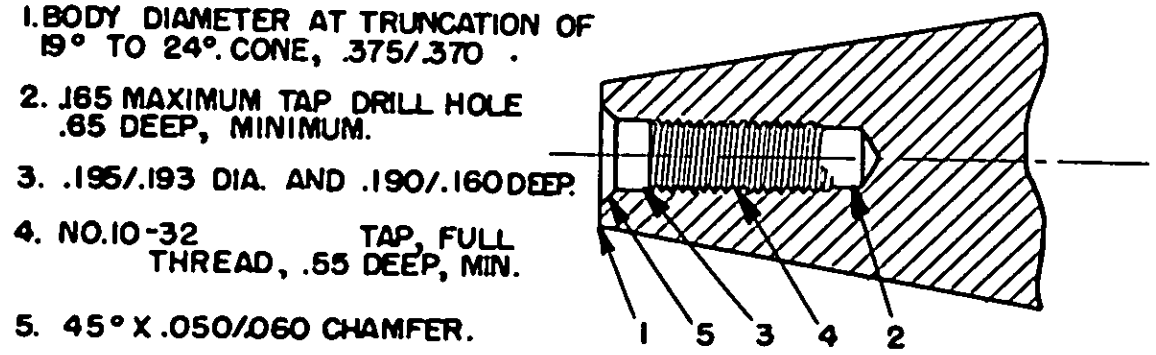


FIGURE 2.—Dimensions in body for receiving point.

3.6.2 *Removable head.* The removable head shall be of brass, properly threaded to fit the body. The design of the head shall be such as to insure that the axis of the plumb bob will hang in the axis of the suspension cord. A spool or other similar device, rotatable under suitable friction control, may be incorporated in the head to provide for precision height adjustment.

3.6.3 *Replaceable points.* Unless a nonmagnetic material is specified (see 6.2 and 6.2.1) replaceable points shall be of steel. Each steel point shall be properly hardened throughout, or for a distance extending approximately 1/4 inch from the tip of the point, in order to conform to the drop test as specified in 4.4.3. The points shall be machined in accordance with dimensions and tolerances specified in figure 3. The points shall have a smooth machine finish (maximum surface roughness of 125 microinches (r.m.s.) and shall be free from fins and burrs. A hole perpendicular to the axis (see figure 3) shall be provided to facilitate insertion and removal of the points. The allow-

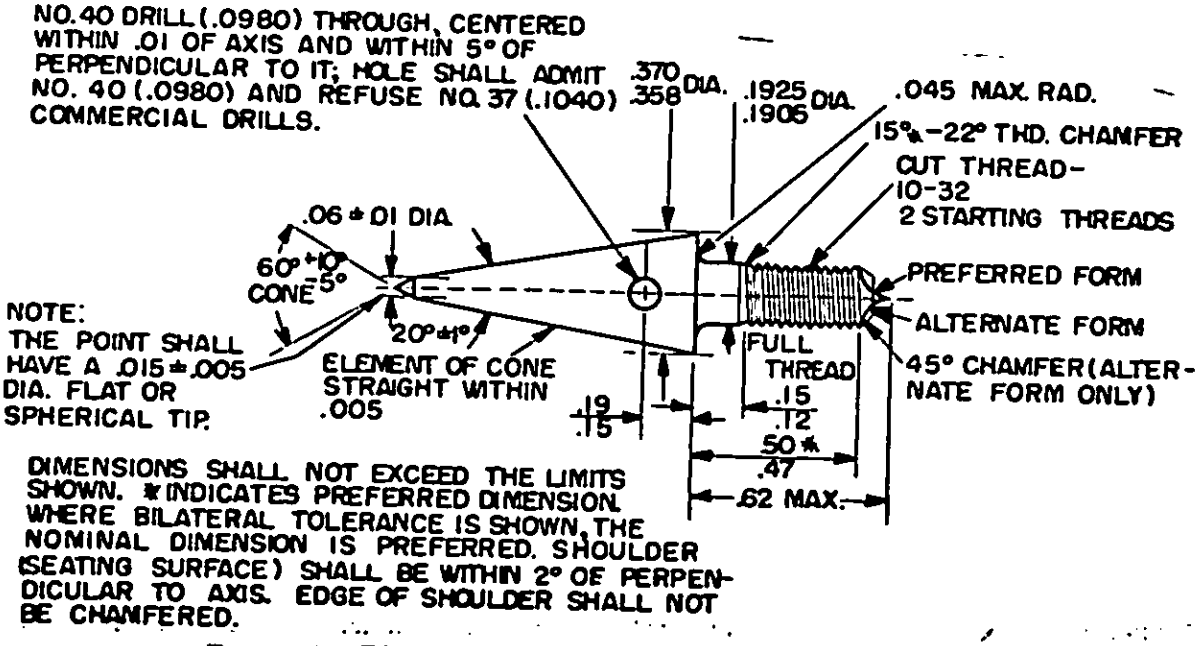


FIGURE 3.—Dimensions for points for type I plumb bobs.

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able flatness on the point shall be in accordance with figure 3.

3.6.4 *Finish.* The head and body shall be polished and lacquered. The points shall have either a natural finish or that resulting from heat treatment (free of scale), or a phosphate coating or other suitable finish including plating with a corrosion resistant metal.

3.6.5 *Concentricity.* Type I plumb bobs shall be concentric within 0.010-inch-total-indicator reading when tested as specified in 4.4.1.

3.6.6 *Replacement points.* The contractors supplying type I plumb bobs shall make available and shall furnish replacement points when requested. All points shall conform to the requirements specified in 3.6.3 in order that points may be interchangeable and replaced without requiring modification. Points shall be concentric within 0.005-inch-total-indicator reading when tested as specified in 4.4.2.

3.7 **Type II plumb bobs, iron, cast.** Type II plumb bobs shall be of solid cast iron, with the head, body, and point cast integrally, shall be similar to figure 4, and conform to table II. The included angle of taper shall not exceed 60 degrees.

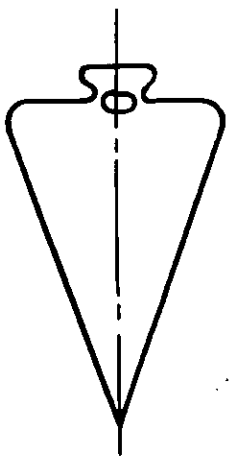


FIGURE 4.—Type II, plumb bob, iron, cast.

Table II - Type II plumb bobs, iron, cast.

Size weight nominal ±10 percent	Diameter of bob at largest section	
	Maximum	Minimum
Ounces	Inches	Inches
9	1-7/16	1-5/16
18	2	1-3/4
32	2-1/2	2-1/4

3.7.1 *Head.* The head shall be provided with two intersecting holes, one through the axis of the head to accommodate the suspension cord, and the other at right angles to the axis to facilitate securing the cord.

3.7.2 *Finish.* Plumb bobs shall be either electroplated with zinc or nickel, japanned or provided with a chemically produced oxide or phosphate coating.

3.7.3 *Concentricity.* Type II plumb bobs shall be concentric within 1/16-inch-total-indicator reading when tested as specified in 4.4.1.

3.7.4 *Point.* The point on all type II plumb bobs shall have a flat or spherical tip not to exceed 3/32-inch diameter.

3.8 **Type III, steel plumb bobs.**

3.8.1 *Class A, mercury-filled, with replaceable points.* Class A plumb bobs shall consist essentially of a body, a removable head, and a replaceable point. A braided nylon or linen cord having a tensile strength of not less than 10 pounds shall be provided with each plumb bob. The length of cords shall be as specified in table III. Type III, class A plumb bobs shall be similar to figure 5, and shall conform to table III.

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Table III - Type III. class A plumb bobs, steel, mercury-filled.

Size weight nominal ±10 percent	Overall length		Diameter of bob at largest section		Length of braided cord
	Maximum	Minimum	Maximum	Minimum	Minimum
Ounces	Inches	Inches	Inches	Inch	Feet
1/3-1/2 [1]	5	4	9/16	1/2	6
6	5-1/2	4-1/2	11/16	9/16	7
8	6	4-3/4	3/4	5/8	8
12	6-1/2	5-3/8	1	7/8	10
16	7	5-7/8	1- 1/8	1	12

[1] Plus 1/2 oz.; minus 10 percent.



FIGURE 5.—Type III, class A, plumb bob, steel, mercury-filled.

3.8.1.1 *Body.* The body shall be provided with an axial cavity containing mercury and shall be fully sealed. The upper portion of the body shall be threaded to receive the removable head. The lower portion of the body shall have an included angle of taper not exceeding 45 degrees, and shall be machined to receive the replaceable point. The periphery of the body shall be suitably knurled for a distance of at least 1/3 the length of the straight portion of the body.

3.8.1.2 *Removable head.* The removable head shall be properly threaded to fit the body. Two intersecting holes shall be provided; one through the axis of the head to accommodate the suspension cord, and the other at right angles to the axis to facilitate securing the cord; or a slotted neck such that the cord may be fastened without employing the use of a knot. The cord hole or slot shall conform to 3.4. The design of the head shall be such as to insure that the axis of the plumb bob will hang in the axis of the suspension cord.

3.8.1.3 *Replaceable points.* Replaceable points shall be provided with a transverse hole to facilitate insertion and removal from the body. The diameter at the largest section of the point shall be approximately the same as the diameter at the smallest section of the body. The point shall terminate in a flat or spherical tip having a diameter between 0.015 and 0.025 inch. All points shall be machined in accordance with good commercial practice. Each point shall be properly hardened throughout, or for a distance extending approximately 1/4 inch from the tip end of the point, in order to conform to the drop test as specified in 4.4.3.

3.8.1.4 *Finish.* Plumb bobs shall be either electroplated with zinc, chrome, nickel, or provided with a chemically produced oxide or phosphate coating.

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3.8.1.5 *Concentricity.* Type III, class A plumb bobs shall be concentric within 0.015-inch-total-indicator reading when tested as specified in 4.4.1.

3.8.1.6 *Replacement points.* The contractors supplying type III, class A plumb bobs shall make available and furnish replacement points when requested. All points shall conform to the requirements specified in 3.8.1.3. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable and replaced without requiring modification. Points shall be concentric within 0.005-inch-total-indicator reading when tested as specified in 4.4.2.

3.8.2 *Type III, class B, plumb bobs, steel, solid.* Class B plumb bobs shall be solid steel with the head, body, and point machined integrally (see A of figure 6) or provided with a removable head (see B of figure 6) or provided with a removable head and replaceable point (see B of figure 6). Plumb bobs shall be similar to A, B, or C of figure 6 and comply with table IV. The included angle of taper shall not exceed 45 degrees.

3.8.2.1 *Body.* The body shall be either round or hexagonal in cross section. Bodies comprising removable heads or removable heads and replaceable points shall be properly threaded to receive these heads and points.

3.8.2.2 *Head.* If the integral head is furnished, it shall be provided with two intersecting holes; one through the axis of the head to accommodate the suspension cord and the other at right angles to the axis to facilitate securing the cord. If the removable head is furnished, it shall be provided with a hole through its axis, or a slotted neck, to accommodate the suspension cord. The cord hole or slot shall conform to 3.4. The design of the head shall be such as to

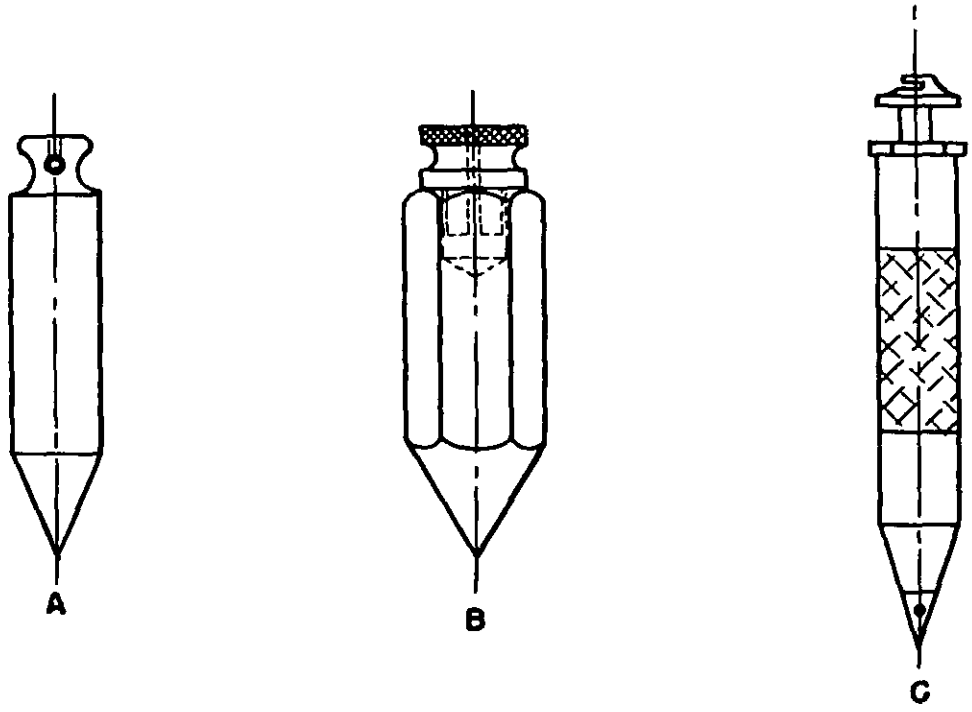


FIGURE 6.—Type III, class B plumb bobs, steel, solid.

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Table IV - Type III, class B plumb-bobs, steel, solid.

Size weight nominal ±10 percent	Overall length		Body			
	Maximum	Minimum	Round diameter		Hexagonal width across flats	
			Maximum	Minimum	Maximum	Minimum
Ounces	Inches	Inches	Inches	Inch	Inches	Inches
[1] 2-1/2	5	2-1/2	5/8	1/2	7/8	5/8
5	5-1/2	3	1-1/16	9/16	7/8	3/4
8	6	4	1-3/8	3/4	1	3/4
12	6	4	1-1/8	7/8	1	7/8
14	6-1/2	4-1/2	1-3/16	1	1-1/8	1
18	6-1/2	4-1/2	1-5/16	1	1-1/4	1-1/8

[1] Plus 1/2 oz.; minus 10 percent.

insure that the axis of the plumb bob will hang in the axis of the suspension cord.

3.8.2.3 *Replaceable points.* When plumb bobs with replaceable points are provided these replaceable points shall conform to the requirements of 3.8.1.3.

3.8.2.4 *Finish.* Plumb bobs shall be either electroplated with zinc, cadmium, chrome, nickel, or provided with a chemically produced oxide or phosphate coating.

3.8.2.5 *Concentricity.* Type III, class B plumb bobs shall be concentric within 0.015-inch-total-indicator reading when tested as specified in 4.4.1.

3.8.2.6 *Replacement points.* The contractors supplying type III, class B plumb bobs with replaceable points shall make available and shall furnish replacement points when requested. All points shall conform to the requirements of 3.8.1.3 and 3.8.1.6.

3.9 Workmanship. Workmanship shall be first class in every respect. Edges and corners shall be rounded in conformance with good commercial standards. The plumb bobs shall have no burrs, fins, sharp projections, excessive surface roughness or any other imperfections which may impair their durability and serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 *Inspection lot.* All plumb bobs of the same type, class, and size presented at one time shall be considered a lot for purposes of inspection.

4.2.2 *Sampling for visual and dimensional examination.* A random sample of plumb bobs shall be selected from each lot of material offered for examination of visual and dimensional characteristics in accordance with MIL-STD-105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300 and inspection level I for lots over 300. The Acceptable Quality Level shall be equal to 1.5 percent defective.

4.2.3 *Sampling for quality conformance tests.* A random sample of plumb bobs shall

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be selected from each lot in accordance with MIL-STD-105 at inspection level L8. The Acceptable Quality Level shall be 2.5 percent defective. However, the sample size shall be the number associated with the letter in the table and acceptance number shall be zero until the acceptable quality level permits one or greater.

4.3 Quality conformance inspection.

4.3.1 *Visual and dimensional examination.* Each of the sample plumb bobs selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Examination shall be conducted as specified in table V. Any plumb bob in the sample containing one or more visual or dimensional defects shall not be offered for delivery and if the number of defective plumb bobs in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

TABLE V.—*Classification of Defects in accordance with MIL-STD-105.*

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type, class, and size not as specified.
102	Incomplete, component parts (if applicable) missing, or improperly assembled.
103	Component parts (if applicable) not new, evidence of repaired, used, or reclaimed parts.
104	Body, head or cap, and point material not as specified.
105	Body not free from blow holes, shrink holes, cracks, sharp edges, burrs, not smooth.
106	Point not finished as specified, not free of fins, burrs, or coarse tool marks.
107	Diameter of point at its vertex exceeds specified values.
108	Spool missing, defective, or not controlled by a suitable friction device (as applicable)
109	Head and/or point (if required) not readily removable.
110	Means of attaching the cord missing, defective, or not as specified.
111	Centerline of the cavity in the body not aligned with axis of body within the allowable deviation (as applicable).
112	Mercury cavity not fully sealed (as applicable).
113	Overall length not within maximum or minimum values.
114	Threads broken, torn, not concentric, or not class of fit specified.
Minor:	
201	Finish or coating defective, incomplete, or not as specified.
202	Angle of taper not as specified.
203	Cord missing, damaged, improperly secured, or not the required length (as applicable).
204	Body cross section not as specified.
205	Diameter (or hexagonal width across flats) not as specified.
206	Weight not as required.
207	Marking, manufacturer's name or trademark missing, incorrect, illegible, or not permanent.

4.3.2 *Tests.* Each of the sample plumb bob selected in accordance with 4.2.3 shall be tested in accordance with 4.4.1, 4.4.2, and 4.4.3 to verify compliance with this specification. Any sample plumb bob which does not meet the requirements for these characteristics shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures.

4.4.1 *Concentricity test for plumb bobs.* Samples of each type plumb bob shall be tested for concentricity (see 3.4) between male and female centers on a testing device whose eccentricity does not exceed 0.001 inch total indicator reading. Concentricity requirements for types I, II, and III plumb bobs shall be as follows:

(*a*) Type I samples shall be concentric throughout their entire conical length within 0.010 inch total indicator reading.

(*b*) Type II samples shall be concentric throughout their entire conical length within 1/16 inch total indicator reading.

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(c) Type III samples shall be concentric throughout their entire body length both conical and cylindrical, within 0.015 inch total indicator reading. In the event the male portion of the body is hexagonal, the concentricity reading shall be taken from both the conical portion and each flat of the hexagonal section.

4.4.2 *Concentricity test for replacement points.* Concentricity of sample replacement points shall be determined by holding the entire shank of the point in an appropriate rotating device whose eccentricity does not exceed 0.001 inch total indicator reading. When the sample point is revolved 360 degrees, it shall be concentric throughout its entire conical length within 0.005 inch total indicator reading.

4.4.3 *Drop test for types I and III plumb bobs with replaceable steel points.* Each sample plumb bob selected for testing shall be held by the suspension cord and dropped onto a steel test plate. The cord may be held between the thumb and forefinger, or by other suitable means. In this position, the point of the sample shall be approximately 18 inches above the steel test plate. The cord is released after the plumb bob is steadied so that the point strikes the steel test plate squarely. The test plate shall be of low carbon cold rolled steel as furnished by the mill. Each sample shall be dropped in this manner 5 times and then inspected. Subsequent inspection shall not show any visual evidence of cracks, breaks, spalling, mushrooming, or bending. Any point failing to pass this test shall be rejected. When "replacement" points only are furnished, samples shall be mounted in an applicable plumb bob weighing at least 16 ounces for type I, and 12 ounces for type III, and tested as described above.

4.4.4 *Possible test failures.* Possible test failures are defined as follows:

(1) Drop test (as applicable for replaceable points):

(a) Evidence of cracks, breaks, spalling, mushrooming or bending after test specimen has been subjected to the specified test.

(2) Concentricity (for plumb bob):

(a) Test specimen not within the specified total indicator reading when tested as specified; not concentric.

(3) Concentricity of point test (for replaceable points):

(a) Eccentricity of point exceeds the specified tolerance; not concentric throughout entire conical length within specified tolerance.

4.5 Examination of preparation for delivery. Sample plumb bobs and packages and packs shall be examined to determine compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.* Plumb bobs shall be preserved and packaged in accordance with level A of MIL-H-15424. Preservative compounds shall not be required when surfaces of plumb bobs are protected with a vitreous or plastic coating, prime coated, painted, or constructed of materials that are inherently resistant to corrosion. Unit packaging for those types of plumb bobs shall be in accordance with method III.

5.1.2 *Level C.* Cleaning, preservation, and packaging shall be in accordance with the suppliers commercial practice.

5.2 Packing.

5.2.1 *Level A.* Plumb bobs, packaged as specified (see 6.2) shall be packed in accordance with the level A requirements of MIL-H-15424.

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5.2.2 *Level B.* Plumb bobs, packaged as specified (see 6.2) shall be packed in accordance with the level B requirements of MIL-H-15424.

5.2.3 *Level C.* Plumb bobs, packaged as specified (see 6.2), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the carrier rules and regulations as applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Nonmilitary marking.* Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military marking.* In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The plumb bobs covered by this specification are generally used for establishing perpendicular lines, and for locating any point vertically above or below another point.

6.1.1 Types I and III are intended for use where considerable accuracy is required while type II is used where close accuracy is not required.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class required (1.2).
(*c*) If a nonmagnetic point is required (3.6.3).
(*d*) Whether replacement points for type III, class A plumb bobs are required (3.8.1.6).
(*e*) Whether replacement points for type III, class B plumb bobs with replaceable points are required (3.8.2.6).
(*f*) Applicable levels of preservation, packaging, and packing required (5.1 and 5.2).

6.2.1 *Replacement points.* Procurement document should specify the following:

(*a*) If a nonmagnetic point is required (3.6.3).
(*b*) Quantity of points required.
(*c*) Type and class of plumb bob that replaceable points are for.

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name. Carload minimum weight 30,000 pounds.

Motor:

Tools, hand, not otherwise indexed. Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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